

INS Assignment 1

Comparative Analysis of Classical Encryption Techniques

1. Analyse the computational complexity of implementing the following techniques:

Playfair Cipher

Hill Cipher

Vigenère Cipher

Playfair Cipher:

For key generation we need to create a 5X5 matrix by removing duplicates from the key and filling the remaining space with the rest of the alphabets. It will take $O(n)$ time (n is key length). Next, we divide plaintext into pairs of characters (digraph), that will take $O(m)$ time (m is length of plaintext). For encrypting, finding the position of the pair of characters and replacing will take $O(1)$ constant time, complexity is $O(n)$. Overall, it takes $O(n)$ time for key generation. Encryption and Decryption takes $O(n)$ time.

Hill Cipher:

For creating the matrix it takes $O(k^2)$ time. For key generation we need to find inverse of matrix which takes $O(k^3)$ time (k is matrix dimension). For encryption, for matrix multiplication we multiply each block of k plaintext letters with key matrix and this takes $O(k^2 \cdot m)$ time (m is length of plaintext, each block requires $O(k^2)$ operations. For the modulo operation, it takes $O(k^2)$ time. Overall, if plaintext has m blocks the time complexity for encryption is $O(mk^2)$.

Vigenère Cipher:

For key generation it takes $O(k)$ time (k is length of key). For encryption repeating key to match plaintext length takes $O(n)$ time (n is length of plaintext). For Caesar cipher shift takes $O(1)$ per letter, overall encryption takes $O(n)$ time.

Cipher	Key Generation	Encryption/Decryption
Playfair	$O(k)$	$O(n)$
Hill	$O(k^3)$	$O(mk^2)$
Vigenere	$O(k)$	$O(n)$

2. Provide a mathematical explanation for breaking each cipher using cryptanalysis techniques.

Breaking Playfair Cipher

Known Plaintext Attack: If attacker knows some plaintext-ciphertext pairs, they can infer parts of the 5X5 matrix. For each known digraph (P_1, P_2) and its corresponding ciphertext digraph (C_1, C_2), the attacker can use Playfair rules by reversing them to find positions of

P1,P2,C1 and C2 in the key matrix. By solving the Playfair rules for multiple known pairs, the attacker can get the key.

Breaking the Hill Cipher

Known-Plaintext Attack: If attacker knows k plaintext ciphertext pairs (k is size of matrix) , key matrix K can be determined by solving a system of linear equations. Let P be $k \times k$ matrix of plaintext blocks, C is ciphertext matrix, the equation is $C = K \cdot P \pmod{26}$. If P is invertible (modulo 26), key matrix K can be found by attacker using $K = C \cdot P^{-1} \pmod{26}$

Breaking the Vigenère Cipher

Kasiski Examination: The attacker can identify the repeating patterns in ciphertext and guess the length of key. Distance between repeated sequences of ciphertext is measured and the GCD of these distances is possibly the length of key.

3. **Design and implement a hybrid cipher combining a substitution and a transposition technique, ensuring:
At least 128-bit encryption strength.
Justification of why this hybrid approach is more secure than using the techniques individually.**

Hybrid Cipher Design

The hybrid cipher combines:

1. **Substitution:** AES (Advanced Encryption Standard) for strong substitution.
2. **Transposition:** A custom block-based transposition cipher.

Steps:

1. Generation of key: We need to generate a 128-bit key for substitution. A separate key for transposition cipher (e.g. Permutation order for blocks) is generated.
2. Next step is Encryption. For that we encrypt the plaintext using AES (substitution) and then divide the AES ciphertext into fixed size blocks (e.g. 128bits) and then apply block-level transposition by rearranging the blocks according to the permutation/transposition key defined earlier.
3. For the Decryption process, rearrange the ciphertext blocks back to their original order using the permutation/transposition key. Then decrypt the rearranged block using AES to get back the plaintext.

Justification:

Combining both substitution and transposition makes sure that an attacker needs to break both the layers to get the plaintext. AES encryption obscures the frequency distribution of plaintext, while transposition disrupts any patterns. The hybrid key uses both an AES key as well as a transposition key thereby increasing effective security. AES alone provides 128-bit encryption strength and the transposition key adds additional complexity making brute force attacks computationally infeasible.

4. Submit a detailed report (2000-3000 words) in PDF form only along with its plagiarism report attached that includes:

- a. Cipher design process.**
- b. Encryption and decryption examples.**
- c. Security evaluation of the hybrid cipher.**
- d. Mathematical formulation of the cipher design process.**
- e. No codes are necessary in the report.**

A hybrid cipher combines two or more ciphers to enhance security.

Cipher Design Process

The main goal is to implement a hybrid cipher that will combine both substitution and transposition cipher to achieve 128-bit encryption strength and hence provide enhanced security.

For the Substitution cipher part, I will implement AES (Advanced Encryption Standard) algorithm. I am taking key size as 128 bits and the mode will be Cipher Block Chaining (CBC). I have used AES since it is a trusted symmetric encryption algorithm.

For the Transposition cipher part, I will use Block Level Permutation taking 128-bit block size so as to match the AES block size, the key will be the permutation order for rearranging the blocks. I have used Block Level Transposition since it changes the patterns of ciphertext and hence making pattern analysis much harder to crack. This will add an additional layer of security.

For the Hybrid approach, for encryption process I will apply Advanced Encryption Standard (AES) first and then to it apply the Block Level Transposition. For decryption, need to reverse the transposition and then decrypt it using AES.

The Steps are:

1. Key generation: First generate a 128-bit AES key for the substitution and then define the transposition key.
2. Encryption: First pad the plaintext so it matches the AES block size. Then encrypt the plaintext using AES. Next divide the AES ciphertext into blocks of 128-bits. Then rearrange the blocks based on the random transposition key that is generated.
3. Decryption: First reverse the transposition to get back the original block order. Then decrypt the rearranged ciphertext using AES. Finally remove the padding if added to get the plaintext.

Encryption and decryption examples.

```
Enter text: Hybrid Cipher
Plaintext: Hybrid Cipher
Ciphertext (hex): 3c8b2ece03af6be51e6e697ed81d59e47b1169565f4e2acd8c812321d9d2631b
Decrypted Text: Hybrid Cipher
```

```
Enter text: This is a secret message
Plaintext: This is a secret message
Ciphertext (hex): 5cc052ade1e05cb059329e6538059d99435a054246f5968e71f16cbc261135ac853259ceb379582122e136821cb172fe
Decrypted Text: This is a secret message
```

Security evaluation of the hybrid cipher

The hybrid cipher will provide 128-bit security through AES that is it would take 2^{128} attempts to break the cipher if the attacker tries and uses brute force. This would be computationally not possible currently. Since the cipher combines both substitution (AES) and transposition, if there is an attack the attacker will need to break both layers to get the plaintext. Even if an attacker knows the transposition key, he/she will still need to break The AES to continue and get the plaintext. AES encryption removes patterns in the plaintext hence preventing attackers trying to use frequency analysis while transposition disrupts any patterns. Without the AES key and knowing only parts of plaintext it is not possible to get full key.

A limitation of this hybrid cipher is the transposition key depends on number of blocks. For n blocks, there are $n!$ keys so if n is very small it is possible to get the key using brute force. Padding in AES may leak information about plaintext length. To help overcome the limitations we can use a large number of blocks to increase the transposition key space and randomized padding scheme can be added.

Mathematical formulation of the cipher design process

AES Encryption:

Let P be plaintext, K be AES key and C the ciphertext. The AES key will be given as:

$$C = \text{AES}_K(P)$$

Transposition:

Let C be AES ciphertext divided into n blocks, that will be $C = [C_1, C_2, C_3, \dots, C_n]$ and let π be the transposition key. The transposed ciphertext C' will be given as

$$C' = [C_{\pi(1)}, C_{\pi(2)}, C_{\pi(3)}, \dots, C_{\pi(n)}]$$

Hybrid Cipher:

The hybrid cipher will combine the AES and transposition giving mathematical formula:

$$C' = \text{Transpose}_{\pi}(\text{AES}_K(P))$$

The decryption process for the hybrid cipher will be given by reversing the transposition that is $C = \text{ReverseTranspose}_{\pi}(C')$ and then decryption using AES is done to get the plaintext.

$$P = \text{AES}_K^{-1}(C)$$

